THE DERIVED HOM IS COMPUTED BY RESOLVING AND APPLYING HOM IN THE DG CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-hom-derived (source lines 2849-2867); the proof environment begins two lines below it and is its own proof, closing at source line 2892.

Editorial note. The proof invokes author results of the same chapter by label (a resolution lemma and a property (P) lemma), retained as context spans, and the generator expands that context transitively; reference targets that live in other chapters but carry canonical project tags are auto-declared as non-load-bearing source pointers, which is what makes this card buildable under the v2 rule.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-dga/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Differential Graded Algebra chapter, counted once: the right derived Hom of dg modules exists and is computed from a property (P) resolution.

Allowed standard prerequisites (editorial).

- Dg modules over a dg algebra and the category K(Mod). The dg category of dg modules and the homotopy category obtained by identifying homotopic morphisms.. The construction and its compatibility with the homotopy category are the setting (source lines 2849-2867, 2875-2892).
- Resolution of a dg module. Every dg module is quasi-isomorphic to one with property (P), so the derived functor can be computed from such a resolution (the author's resolution lemma, retained as context).. It supplies the replacement used to define the derived functor (source lines 2869-2880).
- Homotopy invariance of the Hom complex. The Hom complex depends only on the quasi-isomorphism classes involved (the author's property (P) lemma, retained as context).. It shows the construction does not depend on the chosen resolution (source lines 2880-2890).

Source reference. Original lemma 0149: sources/stacks-chow/tags.txt, source lines 1464–1464. Cross-chapter pointer whose target carries the canonical tag 0149 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 09IZ: sources/stacks-chow/tags.txt, source lines 12057–12057. Cross-chapter pointer whose target carries the canonical tag 09IZ in

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Date: stacks-project:dga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 09J0: sources/stacks-chow/tags.txt, source lines 12058–12058. Cross-chapter pointer whose target carries the canonical tag 09J0 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 014C: sources/stacks-chow/tags.txt, source lines 1467–1467. Cross-chapter pointer whose target carries the canonical tag 014C in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 013N: sources/stacks-chow/tags.txt, source lines 1442–1442. Cross-chapter pointer whose target carries the canonical tag 013N in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Lemma 0.1. *Let (A, d) be a differential graded algebra. Let P be a differential graded A-module with property (P). Then*

$$\mathrm{Hom}_{K(Mod_{(A,d)})}(P, N) = 0$$

for all acyclic differential graded A-modules N.

Proof. We will use that $K(\mathrm{Mod}_{(A,\mathrm{d})})$ is a triangulated category (Proposition 0.7). Let $F_\bullet$ be a filtration on P as in property (P). The short exact sequence of Lemma 0.6 produces a distinguished triangle. Hence by Derived Categories, Lemma 0149 it suffices to show that

$$\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(F_i P, N) = 0$$

for all acyclic differential graded A-modules N and all i. Each of the differential graded modules $F_i P$ has a finite filtration by admissible monomorphisms, whose graded pieces are direct sums of shifts $A[k]$. Thus it suffices to prove that

$$\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(A[k], N) = 0$$

for all acyclic differential graded A-modules N and all k. This follows from Lemma 0.5. □

Lemma 0.2. *Let (A, d) be a differential graded algebra. Let I be a differential graded A-module with property (I). Then*

$$\mathrm{Hom}_{K(Mod_{(A,d)})}(N, I) = 0$$

for all acyclic differential graded A-modules N.

Proof. We will use that $K(\mathrm{Mod}_{(A,\mathrm{d})})$ is a triangulated category (Proposition 0.7). Let $F_\bullet$ be a filtration on I as in property (I). The short exact sequence of Lemma 0.9 produces a distinguished triangle. Hence by Derived Categories, Lemma 0149 it suffices to show that

$$\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(N, I/F_i I) = 0$$

for all acyclic differential graded A-modules N and all i. Each of the differential graded modules I/F_iI has a finite filtration by admissible monomorphisms, whose graded pieces are products of $A^\vee[k]$. Thus it suffices to prove that

$$\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(N, A^\vee[k]) = 0$$

for all acyclic differential graded A-modules N and all k. This follows from Lemma 0.8 and the fact that $(-)^\vee$ is an exact functor. □

Lemma 0.3. *Let (A, d) be a differential graded algebra. Let M be a differential graded A-module. There exists a homomorphism $P \to M$ of differential graded A-modules such that*

(1) *$P \to M$ is a quasi-isomorphism, and*
(2) *P has property (P).*

Proof. Set $M = M_0$. We inductively choose short exact sequences

$$0 \to M_{i+1} \to P_i \to M_i \to 0$$

where the maps $P_i \to M_i$ are chosen as in Lemma 0.10. This gives a "resolution"

$$\ldots \to P_2 \xrightarrow{f_2} P_1 \xrightarrow{f_1} P_0 \to M \to 0$$

Then we set

$$P = \bigoplus\nolimits_{i\geq 0} P_i$$

as an A-module with grading given by $P^n = \bigoplus_{a+b=n} P^b_{-a}$ and differential (as in the construction of the total complex associated to a double complex) by

$$\mathrm{d}_P(x) = f_{-a}(x) + (-1)^a \mathrm{d}_{P_{-a}}(x)$$

for $x \in P^b_{-a}$. With these conventions P is indeed a differential graded A-module. Recalling that each P_i has a two step filtration $0 \to P_i' \to P_i \to P_i'' \to 0$ we set

$$F_{2i}P = \bigoplus\nolimits_{i\geq j\geq 0} P_j \subset \bigoplus\nolimits_{i\geq 0} P_i = P$$

and we add P'_{i+1} to $F_{2i}P$ to get F_{2i+1}. These are differential graded submodules and the successive quotients are direct sums of shifts of A. By Lemma 0.11 we see that the inclusions $F_iP \to F_{i+1}P$ are admissible monomorphisms. Finally, we have to show that the map $P \to M$ (given by the augmentation $P_0 \to M$) is a quasi-isomorphism. This follows from Homology, Lemma 09IZ. □

Lemma 0.4. *Let (A, d) be a differential graded algebra. Let M be a differential graded A-module. There exists a homomorphism $M \to I$ of differential graded A-modules such that*

(1) *$M \to I$ is a quasi-isomorphism, and*
(2) *I has property (I).*

Proof. Set $M = M_0$. We inductively choose short exact sequences

$$0 \to M_i \to I_i \to M_{i+1} \to 0$$

where the maps $M_i \to I_i$ are chosen as in Lemma 0.12. This gives a "resolution"

$$0 \to M \to I_0 \xrightarrow{f_0} I_1 \xrightarrow{f_1} I_1 \to \ldots$$

Denote I the differential graded A-module with graded parts

$$I^n = \prod\nolimits_{i\geq 0} I_i^{n-i}$$

and differential defined by

$$\mathrm{d}_I(x) = f_i(x) + (-1)^i \mathrm{d}_{I_i}(x)$$

for $x \in I_i^{n-i}$. With these conventions I is indeed a differential graded A-module. Recalling that each I_i has a two step filtration $0 \to I_i' \to I_i \to I_i'' \to 0$ we set

$$F_{2i}I^n = \prod\nolimits_{j \geq i} I_j^{n-j} \subset \prod\nolimits_{i \geq 0} I_i^{n-i} = I^n$$

and we add a factor I_{i+1}' to $F_{2i}I$ to get $F_{2i+1}I$. These are differential graded submodules and the successive quotients are products of shifts of $A^\vee$. By Lemma 0.13 we see that the inclusions $F_{i+1}I \to F_iI$ are admissible monomorphisms. Finally, we have to show that the map $M \to I$ (given by the augmentation $M \to I_0$) is a quasi-isomorphism. This follows from Homology, Lemma 09J0. □

Lemma 0.5. *Let (A, d) be a differential graded algebra. Then we have*

$$\mathrm{Hom}_{Mod_{(A,d)}}(A[k], M) = \mathrm{Ker}(d : M^{-k} \to M^{-k+1})$$

and

$$\mathrm{Hom}_{K(Mod_{(A,d)})}(A[k], M) = H^{-k}(M)$$

for any differential graded A-module M.

Proof. Immediate from the definitions. □

Lemma 0.6. *Let (A, d) be a differential graded algebra. Let P be a differential graded A-module. If $F_\bullet$ is a filtration as in property (P), then we obtain an admissible short exact sequence*

$$0 \to \bigoplus F_iP \to \bigoplus F_iP \to P \to 0$$

of differential graded A-modules.

Proof. The second map is the direct sum of the inclusion maps. The first map on the summand F_iP of the source is the sum of the identity $F_iP \to F_iP$ and the negative of the inclusion map $F_iP \to F_{i+1}P$. Choose homomorphisms $s_i : F_{i+1}P \to F_iP$ of graded A-modules which are left inverse to the inclusion maps. Composing gives maps $s_{j,i} : F_jP \to F_iP$ for all $j > i$. Then a left inverse of the first arrow maps $x \in F_jP$ to $(s_{j,0}(x), s_{j,1}(x), \ldots, s_{j,j-1}(x), 0, \ldots)$ in $\bigoplus F_iP$. □

Proposition 0.7. *Let (A, d) be a differential graded algebra. The homotopy category $K(Mod_{(A,d)})$ of differential graded A-modules with its natural translation functors and distinguished triangles is a triangulated category.*

Proof. We know that $K(\mathrm{Mod}_{(A,\mathrm{d})})$ is a pre-triangulated category. Hence it suffices to prove TR4 and to prove it we can use Derived Categories, Lemma 014C. Let $K \to L$ and $L \to M$ be composable morphisms of $K(\mathrm{Mod}_{(A,\mathrm{d})})$. By Lemma 0.14 we may assume that $K \to L$ and $L \to M$ are admissible monomorphisms. In this case the result follows from Lemma 0.15. □

Lemma 0.8. *Let (A, d) be a differential graded algebra. Then we have*

$$\mathrm{Hom}_{Mod_{(A,d)}}(M, A^\vee[k]) = \mathrm{Ker}(d : (M^\vee)^k \to (M^\vee)^{k+1})$$

and

$$\mathrm{Hom}_{K(Mod_{(A,d)})}(M, A^\vee[k]) = H^k(M^\vee)$$

as functors in the differential graded A-module M.

Proof. This is clear from the discussion above. □

Lemma 0.9. *Let (A, d) be a differential graded algebra. Let I be a differential graded A-module. If $F_\bullet$ is a filtration as in property (I), then we obtain an admissible short exact sequence*

$$0 \to I \to \prod I/F_i I \to \prod I/F_i I \to 0$$

of differential graded A-modules.

Proof. Omitted. Hint: This is dual to Lemma 0.6. □

Lemma 0.10. *Let (A, d) be a differential graded algebra. Let M be a differential graded A-module. There exists a homomorphism $P \to M$ of differential graded A-modules with the following properties*

1. *$P \to M$ is surjective,*
2. *$\mathrm{Ker}(d_P) \to \mathrm{Ker}(d_M)$ is surjective, and*
3. *P sits in an admissible short exact sequence $0 \to P' \to P \to P'' \to 0$ where P', P'' are direct sums of shifts of A.*

Proof. Let P_k be the free A-module with generators x, y in degrees k and $k+1$. Define the structure of a differential graded A-module on P_k by setting $\mathrm{d}(x) = y$ and $\mathrm{d}(y) = 0$. For every element $m \in M^k$ there is a homomorphism $P_k \to M$ sending x to m and y to $\mathrm{d}(m)$. Thus we see that there is a surjection from a direct sum of copies of P_k to M. This clearly produces $P \to M$ having properties (1) and (3). To obtain property (2) note that if $m \in \mathrm{Ker}(\mathrm{d}_M)$ has degree k, then there is a map $A[k] \to M$ mapping 1 to m. Hence we can achieve (2) by adding a direct sum of copies of shifts of A. □

Lemma 0.11. *Let (A, d) be a differential graded algebra. Let $M \to P$ be a surjective homomorphism of differential graded A-modules. If P is projective as a graded A-module, then $M \to P$ is an admissible epimorphism.*

Proof. This is immediate from the definitions. □

Lemma 0.12. *Let (A, d) be a differential graded algebra. Let M be a differential graded A-module. There exists a homomorphism $M \to I$ of differential graded A-modules with the following properties*

1. *$M \to I$ is injective,*
2. *$\mathrm{Coker}(d_M) \to \mathrm{Coker}(d_I)$ is injective, and*
3. *I sits in an admissible short exact sequence $0 \to I' \to I \to I'' \to 0$ where I', I'' are products of shifts of $A^\vee$.*

Proof. We will use the functors $N \mapsto N^\vee$ (from left to right differential graded modules and from right to left differential graded modules) constructed in Section 1 and all of their properties. For every $k \in \mathbf{Z}$ let Q_k be the free left A-module with generators x, y in degrees k and $k+1$. Define the structure of a left differential graded A-module on Q_k by setting $\mathrm{d}(x) = y$ and $\mathrm{d}(y) = 0$. Arguing exactly as in the proof of Lemma 0.10 we find a surjection

$$\bigoplus\nolimits_{i \in I} Q_{k_i} \longrightarrow M^\vee$$

of left differential graded A-modules. Then we can consider the injection

$$M \to (M^\vee)^\vee \to (\bigoplus\nolimits_{i \in I} Q_{k_i})^\vee = \prod\nolimits_{i \in I} I_{k_i}$$

where $I_k = Q^\vee_{-k}$ is the "dual" right differential graded A-module. Further, the short exact sequence $0 \to A[-k-1] \to Q_k \to A[-k] \to 0$ produces a short exact sequence $0 \to A^\vee[k] \to I_k \to A^\vee[k+1] \to 0$.

The result of the previous paragraph produces $M \to I$ having properties (1) and (3). To obtain property (2), suppose $\overline{m} \in \mathrm{Coker}(\mathrm{d}_M)$ is a nonzero element of degree k. Pick a map $\lambda : M^k \to \mathbf{Q}/\mathbf{Z}$ which vanishes on $\mathrm{Im}(M^{k-1} \to M^k)$ but not on m. By Lemma 0.8 this corresponds to a homomorphism $M \to A^\vee[k]$ of differential graded A-modules which does not vanish on m. Hence we can achieve (2) by adding a product of copies of shifts of $A^\vee$. □

Lemma 0.13. *Let (A, d) be a differential graded algebra. Let $I \to M$ be an injective homomorphism of differential graded A-modules. If I is graded injective, then $I \to M$ is an admissible monomorphism.*

Proof. This is immediate from the definitions. □

Lemma 0.14. *Let (A, d) be a differential graded algebra. Let $L_1 \to L_2 \to \ldots \to L_n$ be a sequence of composable homomorphisms of differential graded A-modules. There exists a commutative diagram*

$$\begin{array}{ccccccc} L_1 & \longrightarrow & L_2 & \longrightarrow & \ldots & \longrightarrow & L_n \\ \uparrow & & \uparrow & & & & \uparrow \\ M_1 & \longrightarrow & M_2 & \longrightarrow & \ldots & \longrightarrow & M_n \end{array}$$

in $\mathit{Mod}_{(A,d)}$ such that each $M_i \to M_{i+1}$ is an admissible monomorphism and each $M_i \to L_i$ is a homotopy equivalence.

Proof. The case $n = 1$ is without content. Lemma 1.1 is the case $n = 2$. Suppose we have constructed the diagram except for M_n. Apply Lemma 1.1 to the composition $M_{n-1} \to L_{n-1} \to L_n$. The result is a factorization $M_{n-1} \to M_n \to L_n$ as desired. □

Lemma 0.15. *Let (A, d) be a differential graded algebra. Suppose that $\alpha : K \to L$ and $\beta : L \to M$ are admissible monomorphisms of differential graded A-modules. Then there exist distinguished triangles $(K, L, Q_1, \alpha, p_1, d_1)$, $(K, M, Q_2, \beta \circ \alpha, p_2, d_2)$ and $(L, M, Q_3, \beta, p_3, d_3)$ for which TR4 holds.*

Proof. Say $\pi_1 : L \to K$ and $\pi_3 : M \to L$ are homomorphisms of graded A-modules which are left inverse to α and β. Then also $K \to M$ is an admissible monomorphism with left inverse $\pi_2 = \pi_1 \circ \pi_3$. Let us write Q_1, Q_2 and Q_3 for the cokernels of $K \to L$, $K \to M$, and $L \to M$. Then we obtain identifications (as graded A-modules) $Q_1 = \mathrm{Ker}(\pi_1)$, $Q_3 = \mathrm{Ker}(\pi_3)$ and $Q_2 = \mathrm{Ker}(\pi_2)$. Then $L = K \oplus Q_1$ and $M = L \oplus Q_3$ as graded A-modules. This implies $M = K \oplus Q_1 \oplus Q_3$. Note that $\pi_2 = \pi_1 \circ \pi_3$ is zero on both Q_1 and Q_3. Hence $Q_2 = Q_1 \oplus Q_3$. Consider the commutative diagram

$$\begin{array}{ccccccccc} 0 & \to & K & \to & L & \to & Q_1 & \to & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \to & K & \to & M & \to & Q_2 & \to & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \to & L & \to & M & \to & Q_3 & \to & 0 \end{array}$$

The rows of this diagram are admissible short exact sequences, and hence determine distinguished triangles by definition. Moreover downward arrows in the diagram above are compatible with the chosen splittings and hence define morphisms of triangles

$$(K \to L \to Q_1 \to K[1]) \longrightarrow (K \to M \to Q_2 \to K[1])$$

and

$$(K \to M \to Q_2 \to K[1]) \longrightarrow (L \to M \to Q_3 \to L[1]).$$

Note that the splittings $Q_3 \to M$ of the bottom sequence in the diagram provides a splitting for the split sequence $0 \to Q_1 \to Q_2 \to Q_3 \to 0$ upon composing with $M \to Q_2$. It follows easily from this that the morphism $\delta : Q_3 \to Q_1[1]$ in the corresponding distinguished triangle

$$(Q_1 \to Q_2 \to Q_3 \to Q_1[1])$$

is equal to the composition $Q_3 \to L[1] \to Q_1[1]$. Hence we get a structure as in the conclusion of axiom TR4. □

1. Injective modules and differential graded algebras

Lemma 1.1. *Let (A, d) be a differential graded algebra. Let $\alpha : K \to L$ be a homomorphism of differential graded A-modules. There exists a factorization*

$$K \xrightarrow{\tilde{\alpha}} \tilde{L} \xrightarrow{\pi} L, \qquad K \xrightarrow{\alpha} L$$

in $Mod_{(A,d)}$ such that

1. *$\tilde{\alpha}$ is an admissible monomorphism (see Definition 1.2),*
2. *there is a morphism $s : L \to \tilde{L}$ such that $\pi \circ s = id_L$ and such that $s \circ \pi$ is homotopic to $id_{\tilde{L}}$.*

Proof. The proof is identical to the proof of Derived Categories, Lemma 013N. Namely, we set $\tilde{L} = L \oplus C(1_K)$ and we use elementary properties of the cone construction. □

Definition 1.2. Let (A, d) be a differential graded algebra.

1. A homomorphism $K \to L$ of differential graded A-modules is an *admissible monomorphism* if there exists a graded A-module map $L \to K$ which is left inverse to $K \to L$.
2. A homomorphism $L \to M$ of differential graded A-modules is an *admissible epimorphism* if there exists a graded A-module map $M \to L$ which is right inverse to $L \to M$.
3. A short exact sequence $0 \to K \to L \to M \to 0$ of differential graded A-modules is an *admissible short exact sequence* if it is split as a sequence of graded A-modules.

Lemma 1.3. *Let (A, d) be a differential graded algebra. Let M and N be differential graded A-modules.*

1. *Let $P \to M$ be a P-resolution as in Lemma 0.3. Then*
$$\mathrm{Hom}_{D(A,d)}(M, N) = \mathrm{Hom}_{K(Mod_{(A,d)})}(P, N)$$
2. *Let $N \to I$ be an I-resolution as in Lemma 0.4. Then*
$$\mathrm{Hom}_{D(A,d)}(M, N) = \mathrm{Hom}_{K(Mod_{(A,d)})}(M, I)$$

Proof. Let $P \to M$ be as in (1). Since $P \to M$ is a quasi-isomorphism we see that

$$\mathrm{Hom}_{D(A,\mathrm{d})}(P, N) = \mathrm{Hom}_{D(A,\mathrm{d})}(M, N)$$

by definition of the derived category. A morphism $f : P \to N$ in $D(A, \mathrm{d})$ is equal to $s^{-1}f'$ where $f' : P \to N'$ is a morphism and $s : N \to N'$ is a quasi-isomorphism. Choose a distinguished triangle

$$N \to N' \to Q \to N[1]$$

As s is a quasi-isomorphism, we see that Q is acyclic. Thus $\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(P, Q[k]) = 0$ for all k by Lemma 0.1. Since $\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(P, -)$ is cohomological, we conclude that we can lift $f' : P \to N'$ uniquely to a morphism $f : P \to N$. This finishes the proof.

The proof of (2) is dual to that of (1) using Lemma 0.2 in stead of Lemma 0.1. □